Problem Set 2: Macromolecular Interactions

Due: Sunday, September 13 at 11:59 PM **Value: 25 points** **Format: Five-question problem set**

Purpose

Use chemical structure and intermolecular forces to predict how proteins, lipids, carbohydrates, and nucleic acids behave in cells.

Your task

- Q1 (5): Rank ionic, hydrogen-bond, hydrophobic, and van der Waals interactions by expected sensitivity to water and explain your ranking.
- Q2 (5): Predict how replacing a buried leucine with aspartate could affect protein folding. Name the changed chemical property.
- Q3 (5): Draw a condensation reaction joining two amino acids and label the peptide bond, amino terminus, and carboxyl terminus.
- Q4 (5): A protein loses activity at pH 3 but regains it at pH 7. Propose a reversible molecular explanation and distinguish it from peptide-bond hydrolysis.
- Q5 (5): Interpret the binding data below. Identify which ligand has the strongest apparent affinity and explain what evidence is missing before claiming specificity.

Provided evidence or data

Ligand concentration	Protein bound by A	Protein bound by B
1 micromolar	18%	5%
5 micromolar	56%	22%
20 micromolar	88%	61%

Submit

- One PDF with labeled structures and concise mechanism-based answers.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.